

Exception Flow Specification

EV Charging Station Reservation System Problem Statement #21

This document specifies the exception/error flows for UC-02: Reserve Charging Slot. The flows are based on the existing use-case flow specification and the given requirements.

Use Case	UC-02: Reserve Charging Slot
Primary Actor	EV Driver
Secondary Actor	Payment Gateway (external system)

Exception Flow E-01: Payment Declined

- 7a. Payment Declined — the Payment Gateway declines the payment during reservation.
- 7a1. System displays a "payment failed" message and does not lock the socket.
- 7a2. Driver is prompted to try another payment method.
- 7a3. If payment fails again after 2 attempts, the reservation is cancelled, the socket remains available, and the driver is notified.

Exception Flow E-02: Driver Fails to Plug In Within 10 Minutes (No-Show)

- 8a. The driver fails to plug in within 10 minutes of the reserved slot start time.
- 8a1. System detects that the 10-minute plug-in timer has elapsed with no charging session started.
- 8a2. System automatically releases the locked socket, making it available to other drivers.
- 8a3. System applies a no-show penalty charge to the driver's account and sends a notification (NFR-001).

Outcome

- Payment Declined: the reservation is not completed and the socket remains available.
- No-Show: the reserved socket is released and the no-show penalty is applied.